Frederick Huang

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RESEARCH INTERESTS

Modeling, analysis, and design of functional polymeric materials using machine learning (ML) for structure, property, and processing prediction. Broad topics of interest include machine learning for materials characterization, autonomous experimentation, informatics, and inverse design. Current work is in designing antimicrobial polymeric composites, validating ML predictions for dental composites, and mining literature for legacy materials data.

EDUCATION

University of North Carolina at Chapel Hill and North Carolina State University 2025
B. S., *magna cum laude,* in Biomedical Engineering

RESEARCH EXPERIENCE

Research Associate, University of Texas Health Science Center at San Antonio October 2025–present
Department of Comprehensive Dentistry | Advisor: Kyumin Whang, Ph. D.

- Developed a quaternary ammonium silane functionalized polymer composite incorporated with phosphate mineral fillers to demonstrate antifouling activity against *S. mutans*, *S. sanguinis*, and *C. albicans* with remineralization capability in direct/indirect restorations.
- Synthesized and characterized polymer composite formulations predicted by a collaborator-developed machine learning regression model using mechanical load frames, FT-IR, and colorimetry tests among others within ISO 4049 standards to validate predicted performance and inform iterative model refinement.
- Built a literature-mining pipeline (PyMuPDF, a pretrained YOLOv11 model, LLM-based metadata tagging, and PaddleOCR) to extract textual, tabular, and graphical data from literature, generating a structured dataset of ~290 unique formulations from 32 experimental sources to expand model training data and open-source its use.
- Designed and built a custom curing apparatus, reducing degree of conversion variability in batch sample production.
- Conducted blinded testing on an industry-sponsored project, identifying batch-to-batch manufacturing inconsistencies and reporting findings to the production team for quality review.

This work currently has 1 manuscript (second author) and 2 abstracts (first and second author) in preparation, funded by Pulpdent, Inc, and Largent Health, LLC.

Undergraduate Research Assistant, UNC-Chapel Hill August 2024–May 2025
Joint Department of Biomedical Engineering | Advisor: Jeffrey Macdonald, Ph. D.

- Analyzed heterogeneous LC-MALDI-TOF lipidomic datasets from *R. norvegicus* brain and *E. fuscus* liver tissue using LipidSearch 5, identifying phosphatidylcholine and phosphatidylethanolamine as lipid classes most sensitive to sample preparation conditions.
- Built quantitative visualizations to evaluate data quality across preparation conditions, isolating lipid features with statistically significant degradation signatures.
- Proposed candidate lipids in specimens as features for a lipidomic integrity number (LIN), a data quality metric addressing a gap left by existing RNA/DNA-based quality control standards (RIN, DIN).

This work resulted in 1 abstract, funded by NCSU/UNC's Abrams Scholarship program.

Undergraduate Research Assistant, UNC-Chapel Hill December 2023–August 2024
Division of Pharmacoengineering | Advisor: ZhenZhen Chen, Ph. D.

- Synthesized liposomal drug carriers via thin-film hydration and rotary evaporation, followed by extrusion for size control, and prepared samples for transmission electron microscopy (TEM) imaging to characterize morphology.

TEACHING EXPERIENCE

Teaching Assistant, UNC-Chapel Hill Spring 2025
Materials Science of Biomaterials | Course Intructor: Hayley Richardson, Ph. D.
Joint Department of Biomedical Engineering
Led 2 lab sections (37 total students) of 90-student course for majors. Held office hours for exam preparation and graded assignments/exams.

PRESENTATIONS

Poster Presentations

1. Mathur, S., Paye, K. S., **Huang, F.**, Cole, M. N., Gilbert, M. J., Hederick, M. K., (2025). *Brace Check—Data Collection and Monitoring for Pectus Carinatum Patients*, Joint Department of Biomedical Engineering Senior Design Symposium, Raleigh NC.
2. **Huang, F.**, Arciniega C., Ehrmann B., Macdonald, J., (2025). *Quality Management in Lipidomic Extractions to Develop a Lipidomic Integrity Number*, UNC-Chapel Hill Undergraduate Research Symposium, Chapel Hill NC.

HONORS & AWARDS

Interprofessional Practice and Education Award, *UNC-Chapel Hill* 2025
Awarded through School of Medicine for capstone design work targeting patient needs in physical medicine and rehabilitation.

Abrams Scholarship, *North Carolina State University* 2024
Awarded through College of Engineering to 30-35 Biomedical Engineering students conducting research under faculty guidance.

COMMUNITY SERVICE & LEADERSHIP

Classical Music Instructor, *Musical Empowerment*, UNC-Chapel Hill 2022–2025
Galamian-based curriculum for beginner-intermediate violinists from underserved backgrounds.

Vice President, *Club Ultimate Frisbee*, UNC-Chapel Hill 2022–2025
Finance and Operations, >$60,000 operating budget.

PROFESSIONAL EXPERIENCE

IP/Technology Transfer Intern, UT Health San Antonio 2026

SKILLS AND TECHNICAL COMPETENCIES

- **Characterization:** Contact Angle, Indentation, Mechanical Load Frames, FT-IR, SEM, DSC, DMA
- **Packages:** numpy, pandas, sklearn, matplotlib, seaborn, plotly, scipy
- **Tools:** Jupyter, PlatformIO
- **Software:** Onshape, Fusion360, KiCAD, PyMOL, Chimera, STARe, ImageJ
- **Miscellaneous:** Sewing, sectioning, soldering, SLA/FDM printing, Arduino, ESP32, version control, Linux, LaTeX